

SemEval 2026 task 12: Option-Aware Retrieval and Cross-Encoder Reasoning Framework for Abductive Event Reasoning

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Abstract

Abductive reasoning—the capacity to infer the most plausible explanation from incomplete or noisy observations—remains a significant hurdle for language models that often rely on simple associative patterns. In this paper, we present our framework for the SemEval 2026 Task 12. We propose an Option-Aware Retrieval and Cross-Encoder Reasoning Framework designed to bridge the gap between evidence acquisition and causal inference. Our architecture utilizes a dual-track retrieval strategy that gathers both global background and option-specific evidence, ensuring high recall of decisive clues. This is coupled with a 4-pass independent cross-encoder validation using DeBERTa-v3-large, which isolates individual hypotheses to prevent attention dispersion. Experimental results on the official dataset show that our system achieves a robust test score of 0.8358.

1 Introduction

Abductive reasoning, the process of inferring the most plausible explanation for a set of incomplete or noisy observations, is a cornerstone of human-level causal cognition. While language models have demonstrated remarkable performance in deductive and inductive tasks, they often falter in abductive contexts where relevant evidence is sparse or intermixed with irrelevant information. This limitation is particularly critical because real-world event understanding requires shifting from simple associative patterns to robust causal inference.

In this paper, we address this gap by proposing an Option-Aware Retrieval and Cross-Encoder Reasoning Framework. Despite the maturity of individual retrieval and reading comprehension models, their integration for noisy causal inference remains suboptimal. To overcome this, we introduce an architecture specifically configured for the complexities of abductive reasoning. Our

approach integrates targeted, option-specific evidence gathering with a strictly isolated, 4-pass cross-encoder validation. By forcing independent hypothesis verification within a concentrated context, and coupling this with a score-based logical post-processing filter, our framework prevents the attention-dispersion and probabilistic contradictions common in multi-class machine reading comprehension (MRC) models. Consequently, this separated architecture enables language models to distinguish true causal links from spurious correlations, thereby significantly enhancing the reliability of abductive reasoning in complex real-world scenarios.

2 Related Work

Early work on abductive reasoning has focused mainly on common sense reasoning and employed statistical NLP methods such as Pointwise Mutual Information (PMI), based on datasets such as COPA (Choice of Plausible Alternatives) and extensions that emphasize causal plausibility (Roemmele et al., 2011; Gordon et al., 2012). With the advent of large language models (LLMs), the focus has increasingly shifted to the use of LLMs for causal inference tasks (Ma, 2025; Liu et al., 2025). The ART dataset, created by (Bhagavatula et al., 2020), introduced Abductive NLI (Natural Language Inference) as a discriminative choice task, pinpointing a persistent gap between model and human performance.

In practice, prompted generative LLMs are often adapted to closed-set reasoning tasks, by mapping each candidate to an independent prompt and selecting those with the highest conditional likelihood (Rahmani et al., 2025). However, prompting-based methods can be relatively unstable due to prompt-sensitivity, as small prompt changes may lead to large performance changes (Pecher et al., 2026). By contrast, recent work reports that structured

discriminative models can achieve higher accuracy and efficiency than generative models in certain settings like closed-set classification, as they optimize explicit option-level decision boundaries without autoregressive decoding (Pang et al., 2026).

In parallel, to improve factuality and better process domain-specific queries, retrieval augmentation has been explored to ground LLM-based causal inference in external evidence (Lewis et al., 2021; Arslan et al., 2024). Iterative retrieval-based methods further interleave reasoning and evidence acquisition, reporting improved robustness in multi-hop and distractor-rich settings (Yao et al., 2023; Trivedi et al., 2023). Beyond general knowledge grounding, option-aware dense retrieval in multi-choice query answering improves evidence identification, yielding enhanced retrieval quality and better performance (Singh and Shrivastava, 2025).

While substantial work has contributed to the effectiveness of retrieval-augmented generation (RAG) pipelines (Uhm et al., 2025; Brown et al., 2025), where an external retriever fetches evidence to condition generation, few studies systematically integrate option-aware retrieval with a structured discriminative reasoner tailored for abductive reasoning. Our work bridges this gap by combining bi-encoder retrieval with a finetuned cross-encoder reasoner under isolated option-level decision objectives.

3 Methodology

Constructing causal reasoning system for unconstrained textual data requires navigating a fundamental trade-off between evidence recall and reasoning precision. To address this, we propose a two-stage *Option-Aware Retrieval and Cross-Encoder Reasoning Framework*. Our architecture is explicitly designed to decouple the search phase from the reasoning phase, allowing each component to specialize in its respective role.

As visually shown in Figure 1, our framework is composed of three interconnected modules. The left-most block illustrates the Input Phase, where the original query is decomposed into a global event and four option-specific branches. These branches independently feed into the Retriever Module using all-MiniLM-L12-v2, which utilizes deep self-attention distillation (Wang et al., 2020) and is implemented via the Sentence Transformers framework (Reimers and Gurevych, 2019). The module performs targeted searches to construct a

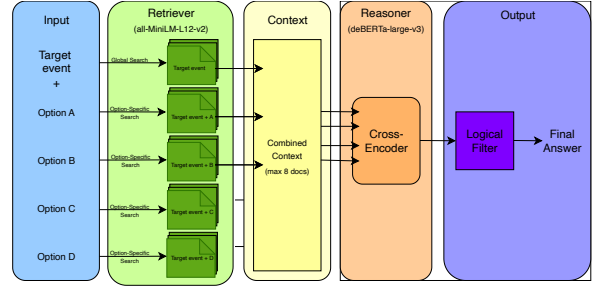


Figure 1: Overview of the proposed Option-Aware Retrieval and Cross-Encoder Reasoning Framework. The pipeline explicitly divides into three stages: (1) extracting dense evidence for the target event and each specific option (Retriever), (2) integrating the retrieved documents into a consolidated context to perform independent 4-pass hypothesis verification (Reasoner), and (3) applying a score-based heuristic to enforce mutually exclusive deterministic outputs (Logical Filter).

highly concentrated, de-duplicated Combined Context (capped at 8 documents). Moving to the right, the Reasoner Module, powered by DeBERTa-v3-large, introduced in (He et al., 2023), receives this context block. Crucially, the figure shows four distinct pathway entering the Cross-Encoder. This indicates our *4-Pass Inference* mechanism, where the model evaluates each candidate causal link sequentially rather than simultaneously. Finally, the four resulting probability scores converge into the Logical Filter block, which resolves any probabilistic contradictions and outputs a single, definitive causality prediction.

3.1 Option-Aware Hybrid Retrieval

If a retrieval system searches documents based solely on the Target Event, it may acquire broad background information but often fails to capture the decisive clues unique to each candidate answer. Based on the insight that a single perspective is insufficient for noisy abductive reasoning, we concluded that relevant evidence is frequently embedded within the specific interaction between the event and its potential causes.

Therefore, we configured our retriever to perform a dual-track search: first it retrieves k_{global} documents using only the Target Event; second, it retrieves an additional k_{option} documents for each Target Event + Option X pair. For the retrieval architecture, we adopted a Bi-Encoder to ensure efficient token-embedding-based search. Finally, the framework filters out redundant documents and concatenates the unique set into a single, comprehensive context.

3.2 Cross-Encoder Reasoning

To maximize precision and validate the candidate causal links identified during the retrieval stage, we employ a high-capacity Cross-Encoder architecture. While Bi-Encoders are efficient for large-scale semantic search, they compute representations for the query and document independently, failing to capture the deep, word-level interactions necessary for complex causal inference. To address this, we utilize DeBERTa-v3-large as our backbone reasoner, allowing the model to perform full-attention across both the hypothesis and the retrieved context simultaneously.

3.2.1 4-Pass Independent Inference

A common approach in MRC is to evaluate all multiple-choice options concurrently using a multi-class softmax layer. However, in noisy abductive reasoning tasks, exposing the model to all candidate options at once often leads to attention dispersion and vulnerability to distractor documents.

To mitigate this, we decompose the multiple-choice problem into four independent binary classification tasks. For each candidate option ($O_i \in \{A, B, C, D\}$), we construct a unified input sequence by concatenating the target event (T), the specific option (O_i), and the consolidated context (C) retrieved from Stage 1:

$$\text{Input}_i = [\text{CLS}] T | O_i [\text{SEP}] C [\text{SEP}]$$

As visually summarized in Figure 2, by isolating each hypothesis, the Cross-Encoder concentrates its entire self-attention mechanism on verifying a single causal narrative at a time. The model outputs a raw logit score (S_i) representing the causal validity of O_i given C . This binary classification format is significantly more robust against the interference of spurious correlations found in distractor options.

3.2.2 Logical Consistency Filter

A fundamental challenge in applying independent binary classifiers to a mutually exclusive task is the emergence of probabilistic contradictions. For instance, the model may confidently assign high probabilities to both a specific causal option (e.g., Option A) and the "None of the others" option (e.g., Option D), resulting in a logical paradox impossible in the ground truth.

To enforce logical consistency without retraining the neural architecture, we implement a *Highest-Score-Wins* heuristic as a definitive post-processing

layer. When a logical conflict is detected among the predicted options (i.e., multiple options including "None" exceed the classification threshold), we bypass the binary thresholding and compare the raw pre-sigmoid logit scores directly:

$$\text{Prediction} = \arg \max_{O_i \in \text{set}} (S_i)$$

This structural logic gate forces the system to commit exclusively to its highest-confidence prediction. By explicitly resolving these neural contradictions, this filter acts as a robust fail-safe, significantly improving the exact-match accuracy of the pipeline and ensuring that the final output strictly adheres to the mutually exclusive constraints of the task.

4 Experimental Setup

4.1 Dataset & Evaluation Metric

We follow the official data split provided by the shared task without using external data. Baseline methods are evaluated on the sample data, as the gold labels for the test set were unavailable before the closure of submission. Models are trained on the training set and finetuned on the development set. Final results are reported on the test set (612 instances) via CodaBench. Each instance consists of a target event and four options as candidate causal hypotheses. Some instances contain a candidate option of the form "None of the others are correct causes" option. Additionally, semantically duplicate options may occur with different option keys. Multiple choices may be correct.

The evaluation logic follows the official metric: an exact match between predicted and gold answer sets receives a score of 1.0; a non-empty subset receives 0.5; all other predictions receive 0. The final score is the mean over all instances.

Furthermore, we assess the system's performance as a ranking task utilizing raw logit scores to order potential candidates. To quantify effectiveness, we employ standard information retrieval metrics, specifically Mean Reciprocal Rank (MRR) and Precision@1 (P@1). These results are then benchmarked against other methods, such as cosine similarity, to analyze the performance-to-complexity trade-offs.

4.2 Main System

4.2.1 Retrieval

We retrieve evidence information using an option-aware, dual-track retrieval strategy. Queries are

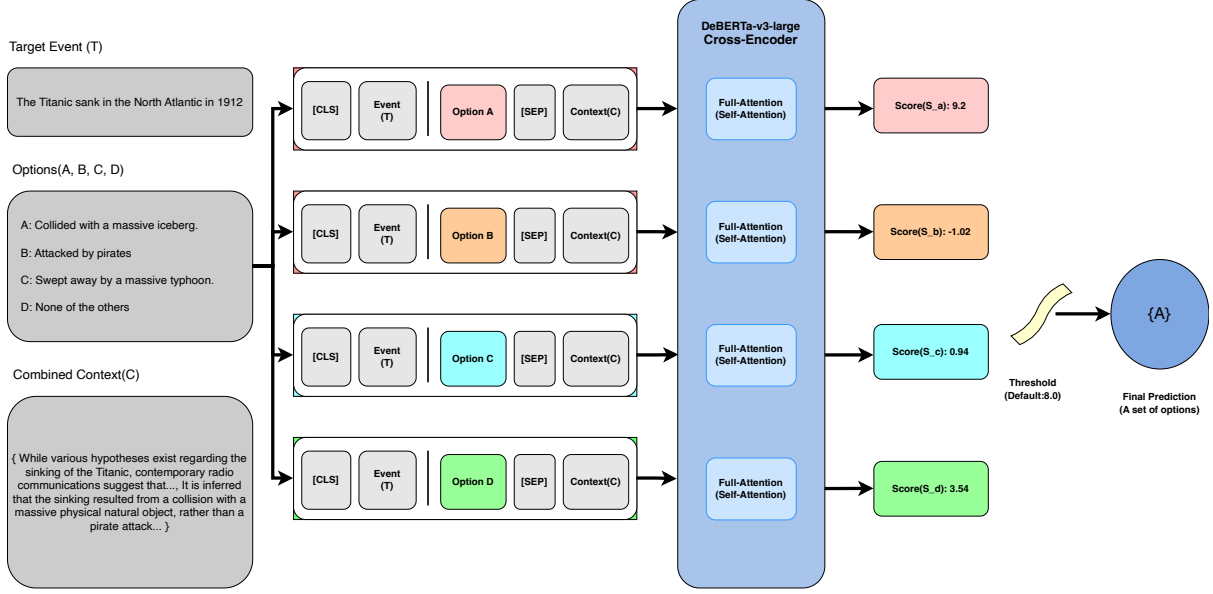


Figure 2: Illustration of the 4-Pass Independent Inference mechanism. As depicted, the single Target Event is paired individually with Option A, B, C, and D. Each pair passes through the DeBERTa Cross-Encoder alongside the Combined Context to produce four distinct raw probability scores (Logits), culminating in a final truth value via the logical filter threshold.

constructed in: (1) the Target Event (global query), and (2) the concatenation of Target Event + Option X pair (option-aware query). Document and query embeddings are computed using the pre-trained bi-encoder all-MiniLM-L12-v2 implemented in SentenceTransformers. Cosine Similarity is used for ranking. We retrieve the top 2 documents from the global query and the top 1 document per option from the option-aware queries. Duplicate documents are removed. Retrieved documents are merged and truncated to typically at most 6 documents per instance; an upper bound of 8 is enforced as a hard constraint on input length.

4.2.2 Training

The reasoner is a cross-encoder initialized from microsoft/deberta-v3-large. The model is trained as a binary classifier over the 4-pass inference mechanism. A candidate receives label 1 if it appears in the gold causal answer set and 0 otherwise. Each training example consists of the option-aware query and the retrieved evidence context. Training is performed for 3 epochs with batch size 4 and learning rate 5×10^{-6} , using AdamW optimization with 300 warm-up steps. Mixed-precision training is enabled. Model selection is based on the development set using option-level binary accuracy with a fixed threshold of 0.5. The best-performing checkpoint is retained.

4.2.3 Inference

At inference time, each option is evaluated independently. Options with scores ≥ 0.5 are selected. If no option exceeds the threshold, the highest-scoring option is selected. If all options exceed the threshold, the lowest-scoring option is removed. Moreover, with the implementation of logic filter, “None of the others” option is only selected when it retains the highest score among the four; semantically duplicate options are constrained to share the identical selection outcome.

5 Supplementary Experiments

Beyond the primary proposed system, we evaluate several baseline architectures to benchmark performance and analyze the trade-off between computational cost and retrieval accuracy. Moreover, to boost the performance of the model, we explored an alternative strategy and an ensemble strategy with LLM-as-judge for adjudication.

5.1 Retrieval Methods

5.1.1 Semantic Vector Search

As a baseline for topical relevance, we implemented a dense retrieval approach using a pre-trained text embedding model. This method calculates the Cosine similarity between document and query embeddings. The underlying hypothesis is that events linked by a causal relationship often

share semantic overlap or exist within similar latent topical spaces.

$$\text{score}(q, d) = \frac{\mathbf{e}_q \cdot \mathbf{e}_d}{\|\mathbf{e}_q\| \|\mathbf{e}_d\|} \quad (1)$$

5.1.2 LLM-Augmented Vector Search

Recent advancements in LLMs have demonstrated significant proficiency in abductive reasoning tasks. Following the approach suggested by Zandie et al. (2023), we leverage the LLM’s internal world knowledge to bridge the gap between an effect and its potential cause during the retrieval phase.

In this configuration, we prepend a generative step to the retrieval pipeline using the following prompt:

“Given the event: '[target event]', what is a plausible cause? Answer concisely in one sentence.”

The resulting generated hypothesis is then embedded and used as the query vector for Cosine similarity search. This “Query Expansion via Reasoning” aims to align the search vector more closely with the semantic space of the target causal documents.

5.1.3 Chunk-level Search

To explore whether an alternative strategy could improve system performance, we replaced the bi-encoder retriever with a chunk-level retriever. By applying semantic chunking with dynamic splits based on cosine similarity, we segmented the context documents into variable-length chunks. We then employed a combination of BM25 for sparse retrieval and BGE-M3 for dense retrieval, whose results were merged via Reciprocal Rank Fusion (RRF) to produce a top-30 candidate pool. A cross-encoder reranker (bge-reranker-v2-m3) then selected top-10 chunks, yielding approximately 300 tokens per option, also a potential causal hypothesis, as input to the DeBERTa reasoner.

5.2 LLM-as-judge Ensemble

After implementing the chunk-level search with the reasoner (deBERTa-v3-large), complementary errors were observed from a comparison with our main system. Following that, we designed a disagreement-driven ensemble strategy. For instances received identical predictions from the main model and the chunk-level-search-based model, we retained the results directly. For samples where the two models disagreed, we submitted

both predictions along with the retrieved evidence to GPT-o3 as an LLM-as-judge for adjudication. The judge was instructed to select all plausible causes based on the evidence, with explicit constraints on the consistent treatment of duplicate options and the conditional use of the “none of the others are causes” option (full prompt provided in Appendix A), aligning with the logical filter design.

6 Results and Analysis

6.1 Overview

We mainly evaluate our system performance using the official metric. Table 1 shows this result for our training set, the validation (dev) set, as well as the released test set.

Table 1: System Performance: Average Scores

Dataset Split	Average Score
Train	0.9164
Dev	0.9750
Test	0.8358

The system achieves a high level of accuracy across all splits, peaking at an average score of 0.9750 on the development set. While the test performance remains robust (0.8358), there is a noticeable delta between the development and test phases that suggests the model may be highly sensitive to the specific distribution of the development set. Which is understandable considering the topic distribution shift. Specifically, the development set shares the same topics as the training set, whereas the test set introduces entirely new topics. The performance drop exhibited on out-of-distribution data indicates limitations in cross-topic generalization.

6.2 Comparison

Table 2: Ranking Performance in comparison with our supplementary experiments. Results shown for test set only

Method	MRR	Precision@1
Semantic Vector Search	0.6844	0.4534
LLM-Augmented Vector Search	0.7327	0.5285
Chunk-level Search + Finetuned deBERTa	0.8764	0.7958
Doc-level Search + Fine-tuned deBERTa	0.9039	0.8420
LLM-as-judge Ensemble	0.9623	0.9412

As described in table 2, the transition from traditional vector-based methods to a transformer-based architecture results in a substantial performance

gain. Specifically, deBERTa outperforms Augmented Cosine Similarity by approximately 17% in MRR and 31% in Precision@1. This indicates that while cosine similarity provides a respectable baseline, it lacks the semantic depth required to accurately identify the top-ranked candidate in more complex scenarios.

It should be noted that the gap is much wider for Precision@1 (0.5285 vs. 0.8420) than for MRR (0.7327 vs. 0.9039). This implies that while cosine similarity can often “get close” (ranking the correct answer in the top 2 to 4), it struggles to consistently place the correct answer at the very top. In contrast, deBERTa demonstrates exceptional precision, making it far more suitable for systems where only the top result is shown to the user.

When replacing the doc-level, option-aware hybrid retrieval strategy (our main architecture) with chunk-level search, the performance exhibits a relatively uniform decrease across both Precision@1 (0.8764 vs. 0.9039) and MRR (0.7958 vs. 0.8420), receiving a score of 0.7132. This could be attributed to a loss of semantic coherence caused by chunking (Merola and Singh, 2025). DeBERTa’s disentangled attention mechanism is particularly sensitive to input coherence (He et al., 2020). Its fine-grained relational reasoning could be undermined when the input is fragmented.

The result of LLM-as-judge ensemble system, by capturing the complementary aspects of the two strategies, receives higher performance across the two metrics (0.9623 vs. 0.9039 for MRR and 0.9412 vs. 0.8429 for Precision@1) with a final score of 0.9020. However, this approach incurs significant workload of implementation overhead and relies on the availability of two independent systems, which makes it less suitable as a standalone pipeline.

6.3 Error Analysis

To understand the limitations of the proposed model, we conducted a qualitative analysis of the instances where the model assigned high confidence scores (> 0.90) to non-golden options, shown in table 3. We identified the following primary failure modes:

- **Lexical Overlap Bias:** The model frequently over-relies on keyword matching. For instance, in question q-2494, the presence of the verb “launched” in both the event and the option triggered a high score, despite the op-

tions describing different stages of the mission (lunar vs. terrestrial launch).

- **Narrative Conflation:** The model struggles with temporal and causal boundaries within the same news cycle. In the context of the Syrian conflict (e.g., q-2610), it conflated the retreat of the army with specific rebel offensives, failing to isolate the unique factual relationship required for the “golden” label.
- **Entity Distraction:** High-profile entities (e.g., *Hezbollah* or *Chang’e 5*) act as anchors. The model often assigns high scores to any option that contains these entities, regardless of the specific action or outcome described in the premise.

7 Conclusion

In this work, we introduced a two-stage retrieval and reasoning framework tailored for the nuances of abductive event reasoning. We demonstrated that option-aware dense retrieval effectively captures the specific interactions between events and their potential causes. While our system achieved high precision and strong benchmarking results on the SemEval 2026 test set, our error analysis highlights a persistent “Lexical Overlap Bias” and a tendency for the model to be distracted by high-profile entities. These findings suggest that while structured discriminative models offer both accuracy over simple vector comparisons and flexibility over generative approaches, future research should focus on enhancing cross-topic generalization and developing mechanisms to better handle temporal and causal boundaries in complex news cycles.

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A LLM-as-Judge Prompt

The following prompt was used for GPT-o3 in the LLM-as-Judge ensemble experiment described in 5.2.

LLM-as-Judge Prompt

You are an expert in causal event reasoning. Your task is to determine which of the following options are plausible causes of the target event, given the retrieved evidence and two model predictions.

Target Event: {target_event}

Retrieved Evidence:
{evidence}

Options:

- A. {option_a}
- B. {option_b}
- C. {option_c}
- D. {option_d}

Model 1 Prediction: {model_1_prediction}

Model 2 Prediction: {model_2_prediction}

Instructions:

1. Based on the retrieved evidence and your own reasoning, determine which options are plausible causes of the target event. Multiple options may be selected.
2. If two or more options have identical content, treat them consistently – either all selected or all rejected.
3. Select “None of the others” only if you are confident that none of options A, B, and C are plausible causes. Do not select it as a default or fallback.
4. Consider both model predictions as references, but make your final judgment based on the evidence and causal plausibility, not by simply deferring to either model.

Respond strictly in the following format:

[option] A, C

[reasoning] {your reasoning here}

B Error Analysis

Table 3: Examples of fail predictions

q-2836	On Dec 5, the Chang’e 5 ascender docked with the return spacecraft.	
Status	Option Text	Predicted Score
Non-Golden	Chang’e 5 separated into the orbiter-reentry capsule combination and the lander-ascender combination.	0.9998
Non-Golden	Chang’e 5 launched on November 23 atop a Long March 5 rocket.	0.9993
Golden	The Chang’e 5 lander gathered rocks and soil from the Moon.	0.0001
Non-Golden	Chang’e-5 raised a Chinese flag on the Moon.	0.9993
q-2494	Chang’e 5 launched from the lunar surface Dec. 3.	
Status	Option Text	Predicted Score
Non-Golden	Chang’e 5 launched on November 23 atop a Long March 5 rocket.	0.9997
Non-Golden	Chang’e 5 mission entered lunar orbit on November 28.	0.0002
Golden	The Chang’e 5 lander gathered rocks and soil from the Moon.	0.0001
Non-Golden	Chang’e-5 raised a Chinese flag on the Moon.	0.0001
q-2610	Syrian army notified officers that Assad’s rule had ended.	
Status	Option Text	Predicted Score
Non-Golden	Rebels take control of Aleppo city on November 30 after an offensive that kills dozens of government soldiers, causing the Syrian army to retreat.	0.9998
Golden	Former Syrian President Bashar al-Assad flees to Russia with his family.	0.0001
Non-Golden	Insurgents entered Saydnaya military prison and freed prisoners.	0.0001
Golden	Former Syrian President Bashar al-Assad flees to Russia with his family.	0.0001
q-2490	Lebanon, Iraq, and Syria declared three days of mourning after the airstrike.	
Status	Option Text	Predicted Score
Non-Golden	Israeli airstrikes on Saturday targeted areas near Beirut and Hezbollah weapons caches, resulting in at least 33 killed and 195 wounded, according to Lebanon’s health ministry.	0.9998
Golden	The airstrike killed Hezbollah Secretary-General Hassan Nasrallah, commander Ali Karaki, additional Hezbollah commanders, and IRGC Brigadier General Abbas Nilforooshan.	0.9998
Non-Golden	Residents fled Dahiyeh after intense Israeli strikes, seeking shelter elsewhere in Beirut.	0.0001
Non-Golden	Prime Minister Benjamin Netanyahu cut short his U.S. visit and ended a UN briefing after being informed of the attack.	0.0001